

Clement Fung

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Education

Carnegie Mellon University, School of Computer Science

Pittsburgh, PA, USA

PH.D IN SOCIETAL COMPUTING, SOFTWARE AND SOCIETAL SYSTEMS DEPARTMENT (GPA: 4.0 / 4.33)

08/2019 - 09/2025

- Thesis: "Proposing Guidelines and Approaches to Make Anomaly Detection More Effective for Industrial Control Systems"
Advisor: Lujo Bauer

University of British Columbia

Vancouver, BC, Canada

M.SC IN COMPUTER SCIENCE (GPA: 88 / 100)

09/2016 - 12/2018

- Thesis: "Dancing in the dark: Private Multi-Party Machine Learning in an Untrusted Setting"
Advisor: Ivan Beschastnikh

University of Waterloo

Waterloo, ON, Canada

B.A.SC IN SYSTEMS DESIGN ENGINEERING, HONOURS (GPA: 88 / 100)

09/2011 - 05/2016

Research

REFEREED PUBLICATIONS

Model Selection of Anomaly Detectors in the Absence of Labeled Validation Data

TAI 2025

Clement Fung, Chen Qiu, Aodong Li, Maja Rudolph.

IEEE Transactions on Artificial Intelligence, Volume 6, Issue 12.

Adopting AI to Protect Industrial Control Systems: Assessing Challenges and Opportunities from the Operators' Perspective

SOUPS 2025

Seattle, WA, USA

Clement Fung, Eric Zeng, Lujo Bauer.

21st Symposium on Usable Privacy and Security.

Targeted Image Transformation for Improving Robustness in Long Range Aircraft Detection

IROS 2024

Abu Dhabi, UAE

Rebecca Martin, Clement Fung, Nikhil Varma Keetha, Lujo Bauer, Sebastian Scherer.

2024 IEEE/RSJ International Conference on Intelligent Robots and Systems.

Attributions for ML-based ICS Anomaly Detection: From Theory to Practice

NDSS 2024

San Diego, CA, USA

Clement Fung, Eric Zeng, Lujo Bauer.

31st Network and Distributed System Security Symposium.

Perspectives from a Comprehensive Evaluation of Reconstruction-based Anomaly Detection in Industrial Control Systems

ESORICS 2022

Copenhagen, Denmark

Clement Fung, Shreya Srinarasi, Keane Lucas, Hay Bryan Phee, Lujo Bauer.

27th European Symposium on Research in Computer Security.

Biscotti: A Blockchain System for Private and Secure Federated Learning

TPDS 2021

Muhammad Shayan, Clement Fung, Chris J.M. Yoon, Ivan Beschastnikh.

IEEE Transactions on Parallel and Distributed Systems, Volume 32, Issue 7.

Towards a Lightweight, Hybrid Approach for Detecting DOM XSS Vulnerabilities with Machine Learning

WWW 2021

Ljubjana, Slovenia (Virtual)

William Melicher, Clement Fung, Lujo Bauer, Limin Jia.

The Web Conference 2021.

The Limitations of Federated Learning in Sybil Settings

RAID 2020

San Sebastian, Spain (Virtual)

Clement Fung, Chris J.M Yoon, Ivan Beschastnikh.

23rd International Symposium on Research in Attacks, Intrusions and Defenses.

Brokered Agreements in Multi-Party Machine Learning

APSys 2019

Hangzhou, China

Clement Fung, Ivan Beschastnikh.

10th ACM SIGOPS Asia-Pacific Workshop on Systems.

GainForest: Scaling Climate Finance for Forest Conservation using Interpretable Machine Learning on Satellite Imagery

ICML 2019 Workshop

Long Beach, CA, USA

David Dao, Catherine Cang, Clement Fung, Ming Zhang, Nick Pawlowski, Reuven Gonzales, Nick Beglinger, Ce Zhang

Climate Change: How Can AI Help?: ICML 2019 Workshop

OTHER PUBLICATIONS

Attacking Autonomous Driving Agents with Adversarial Machine Learning: A Holistic Evaluation with the CARLA Leaderboard

Henry Wong, Clement Fung, Weiran Lin, Karen Li, Stanley Chen, Lujo Bauer.
ArXiv Preprint: 2511.14876

[ArXiv 2025](#)

Dancing in the Dark: Private Multi-Party Machine Learning in an Untrusted Setting

Clement Fung, Jamie Koerner, Stewart Grant, Ivan Beschastnikh.
ArXiv Preprint: 1811.09712

[ArXiv 2018](#)

Mitigating Sybils in Federated Learning Poisoning

Clement Fung, Chris J.M. Yoon, Ivan Beschastnikh.
ArXiv Preprint: 1808.04866

[ArXiv 2018](#)

INVITED TALKS

Mind the Gap: Improving the Effectiveness of Machine Learning for Industrial Control Systems Security

- George Mason University. April 2026.
- University of Maryland Baltimore County. April 2026.
- George Washington University. February 2026.
- DePaul University. January 2026.
- New Jersey Institute of Technology. January 2026.
- CyLab Privacy Seminar. January 2026.

Adopting AI to Protect Industrial Control Systems: Assessing Challenges and Opportunities from the Operators' Perspective

- CyLab Partners Conference. October 2025.
- RSTCON 2025. October 2025.
- Symposium on Usable Privacy and Security (SOUPS). August 2025.

Approaches for More Effective ML-based Anomaly Detection in Industrial Control Systems

- Secure Cyber-Physical Systems Group, CISPA Helmholtz Center for Information Security. October 2024.
- SysNets Seminar, Max Planck Institute for Software Systems. October 2024.
- Artificial Intelligence and Security Group, Karlsruhe Institute of Technology. October 2024.
- CAE-R Research Symposium, National Cybersecurity Education Colloquium. October 2024.

Attributions for ML-based ICS Anomaly Detection: From Theory to Practice

- RSTCON 2025. October 2025.
- Network and Distributed System Security Symposium (NDSS). February 2024.

Perspectives from a Comprehensive Evaluation of Reconstruction-based Anomaly Detection in Industrial Control Systems

- CAE-R Seminar, April 2023.
- European Symposium on Research in Computer Security (ESORICS). September 2022.

Detecting and Explaining Anomalies in Industrial Control

- Bosch Center for Artificial Intelligence. June 2023.
- Accountable Systems Lab, Carnegie Mellon University. November 2022.
- CyLab Partners Conference. October 2022.

Towards a Lightweight, Hybrid Approach for Detecting DOM XSS Vulnerabilities with Machine Learning

- The Web Conference. April 2021.

The Limitations of Federated Learning in Sybil Settings

- Symposium on Research in Attacks, Intrusions and Defenses (RAID). October 2020.

Brokered Agreements in Multi-Party Machine Learning

- Asia-Pacific Workshop on Systems (APSys). August 2019.

Dancing in the Dark: Private Multi-Party Machine Learning in an Untrusted Setting

- UBC Cybersecurity Summit. May 2018.
- Computer Systems Lab Seminar, University of Toronto. December 2017.

Teaching and Mentoring

TEACHING

Carnegie Mellon University

TEACHING ASSISTANT

- 17-331: Information Security, Privacy, and Policy
Instructor: Norman Sadeh
Fall 2024
- 11-667: Large Language Models Methods and Applications
Instructors: Daphne Ippolito, Chenyan Xiong
Fall 2023

FUTURE FACULTY PROGRAM, EBERLY CENTER FOR TEACHING EXCELLENCE AND EDUCATIONAL INNOVATION

- Completed a program that prepares graduate students for teaching in future faculty careers
2024
- Designed a course syllabus, taught two guest lectures, and participated in nine teaching seminars

University of British Columbia

TEACHING ASSISTANT

- DSCI 571: Supervised Learning
Instructors: Michael Gelbart, Varada Kolhatkar
Fall 2018
- DSCI 523: Data Wrangling
Instructors: Jenny Bryan, Rodolfo Lourenzutti
Fall 2018
- CPSC 340: Machine Learning
Instructor: Michael Gelbart
Winter 2018
- CPSC 340: Machine Learning
Instructor: Mark Schmidt
Fall 2017
- CPSC 210: Software Construction
Instructors: Norman Hutchinson, Paul Carter, Mehrdad Oveisi
Winter 2017
- CPSC 210: Software Construction
Instructors: Norman Hutchinson, Ryan Vogt, Jonatan Schroeder
Fall 2016

MENTORING

Carnegie Mellon University

ADVISED RESEARCH ASSISTANTS

- Alexa Lowe
Fall '25, Spring '26
- Denis Wambold
KIT MS Thesis: visited CMU Fall '25
- Henry Wong
Summer '24, Fall '24, Spring '25
- Karen Li
Fall '23, Spring '24
- Stanley Chen
Spring '23
- Sneha Rander
Fall '22
- Jeneel Mashru
Fall '22
- Kevin Li
Fall '22
- Shreya Srinarasi
Summer '21

University of British Columbia

ADVISED RESEARCH ASSISTANTS

- Chris J.M. Yoon
Summer '18, Fall '18
- Syed Mubashir Iqbal
Summer '17
- Jaime Koerner
Spring '17

Professional Experience

Carnegie Mellon University, School of Computer Science

Pittsburgh, PA, USA

POSTDOCTORAL RESEARCH ASSOCIATE

10/2025 - present

- Research on AI-based approaches for securing industrial control systems
- Research on the robustness of ML models for aerial drones

Bosch Center for Artificial Intelligence

Pittsburgh, PA, USA

MACHINE LEARNING RESEARCH INTERN

05/2023 - 08/2023

- Research on applications of diffusion models to improve the performance of visual anomaly detection.
- A paper describing our results was published at IEEE TAI 2025.

Oasis Labs

Berkeley, CA, USA

SOFTWARE ENGINEER

01/2019 - 07/2019

- Developing a secure, private, data-sharing platform for an early-stage blockchain startup.
- A paper describing a use case of this platform was published at an ICML 2019 workshop.

LinkedIn Corporation

Sunnyvale, CA, USA

SOFTWARE ENGINEERING INTERN

06/2015 - 08/2015

- Search, Network, and Analytics Team: Building infrastructure for online ML-based relevance scoring at scale

LinkedIn Corporation

SOFTWARE ENGINEERING INTERN

- Distributed Data Systems Team: Designed and prototyped a derived-data serving system, Venice

Mountain View, CA, USA

09/2014 - 12/2014

Voicebox Technologies

SOFTWARE ENGINEERING INTERN

- Server and Tools Team: Developed concurrent database access layer for a mobile voice-assistant service

Bellevue, WA, USA

01/2014 - 04/2014

Ontario Institute for Cancer Research

SOFTWARE DEVELOPER INTERN

- Developer for the Boutros Lab: a bio-informatics research group

Toronto, ON, Canada

05/2013 - 08/2013

Service

External Service

PROGRAM COMMITTEE

- RAID 2026
- DSML 2026
- VehicleSec 2026
- ACM SoCC 2026
- CPSS 2026
- ISC 2025
- RAID 2025
- WISW 2024
- ACM FAccT 2023
- ACM FAccT 2022

EXTERNAL REVIEWER

- SOUPS 2026
- USENIX Security 2024 (4x)
- IEEE S&P 2024
- IEEE SaTML 2024 (2x)
- USENIX Security 2023 (2x)
- USENIX Security 2022
- IEEE S&P 2021
- NDSS 2021 (2x)
- USENIX Security 2020
- SOUPS 2020
- ISSRE 2017

INVITED REVIEWER

- IEEE Trans. Dependable and Secure Computing 2025
- IEEE Trans. Information Forensics & Security 2025 (2x)
- Springer Machine Learning 2025
- IEEE Trans. Networking 2024
- ACM CCS Posters 2021
- IEEE Trans. Industrial Informatics 2021

CONFERENCE VOLUNTEER

- WOSOC 2024
- SOUPS 2021
- EuroS&P 2021
- EuroS&P 2020

Internal Service

CMU SOFTWARE AND SOCIETAL SYSTEMS DEPARTMENT

- SCS Graduate Application Support Program, 2025
- Student Representative for Faculty Meetings, 2024–2025
- PhD Student Admissions Committee, 2023
- Faculty Hiring Committee, 2022
- Community Building Committee, 2022–2023
- Prospective PhD Student Visit Day Organizer, 2020–2022

CMU CYLAB

- CyLab Sports Committee, 2021–2023

UBC DEPARTMENT OF COMPUTER SCIENCE

- Computer Science Graduate Student Association (President), 2017
- Department Strategy Committee, 2017
- Space and Safety Committee, 2017

Awards and Recognitions

2025	PhD Student Research Award
2024	Best Poster Award (\$300)
2024	AINet Fellow in “Safety and Security of AI”
2017	CS Department Graduate Teaching Assistant Award
2017	CS Department Student Service Award
2016	Sanford Fleming Award for Co-operative Proficiency
2016	GM Canada Innovation Award (\$500)
2015	W.W. King Exchange Fellowship (\$500)
2014	President’s International Experience Award (\$1500)
2013	Sanford Fleming Award for Outstanding Work Term Report (\$300)
2011	Colonel Hugh Heasley Engineering Scholarship (\$10000)
2011	President’s Scholarship of Distinction (\$2000)

RSTCON 2025
IAP CMU Workshop: AI and Security in the Cloud
Deutscher Akademischer Austauschdienst (DAAD)
University of British Columbia
University of British Columbia
University of Waterloo
University of Waterloo
University of Waterloo
University of Waterloo
University of Waterloo
University of Waterloo
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